



An Intelligent IoT and AI-Based Maternal Health Monitoring, Risk Prediction and Emergency Alert System

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Project Title: An Intelligent IoT and AI-Based Maternal Health Monitoring, Risk Prediction and Emergency Alert System

Project Overview:

Materna AI is an intelligent IoT- and AI-based system designed for real-time maternal health monitoring, risk prediction, fall detection, and emergency alerting, particularly for pregnant women in rural and low-resource areas. The system uses an ESP32 integrated with ECG, MAX30105, MPU6050, BMP280, and GPS modules to collect physiological, motion, environmental, and location data. The collected data are transmitted wirelessly to an AI backend for maternal health risk prediction. Six machine learning models, including Logistic Regression, Decision Tree, K-Nearest Neighbors, Gradient Boosting, XGBoost, and Random Forest, were evaluated. Random Forest achieved an accuracy of 93.08% and was selected to classify maternal health risk into Low, Mid, and High Risk categories.

For fall detection, a dataset of 6,446 images was used to evaluate YOLOv8n, YOLOv9t, YOLO11s, and YOLO26n. YOLOv8n achieved 99.7% validation recall, 99.4% mAP@50, and an inference time of 2.3 ms per image, supporting its use for real-time fall detection. The system also includes a 128 $\times$ 64 SSD1306 OLED display, ECG-based buzzer alerts, GPS tracking, Bluetooth communication, and ThingSpeak cloud monitoring. A Flask-based web dashboard supports patient and doctor management, health risk assessment, appointments, consultations, and remote monitoring. The collected data are transmitted to the Flask backend and stored in a MongoDB database for monitoring and analysis. Overall, Materna AI integrates IoT sensing, machine learning, fall detection, and emergency support into a unified platform for continuous maternal health monitoring.

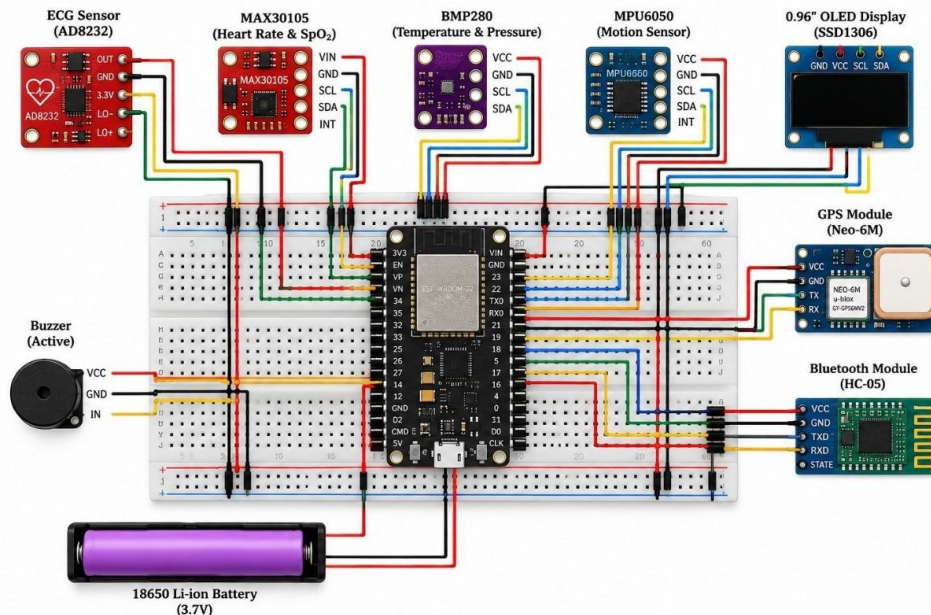
Comparison of Relevant Previous Studies and the Proposed Materna AI System:

Author(s), Year	Method / Model	Technology / Data	Main Contribution	Limitation / Research Gap
Tappari et al., 2026 [1]	Random Forest, IoT	Wearable sensors, ESP32, ThingSpeak	Maternal and child health monitoring with machine learning-based risk assessment	Does not integrate fall detection, GPS tracking, or doctor interaction.
Karvenathan et al., 2026 [2]	IoT-based wearable system	Wearable belt, physiological data, cloud	Continuous maternal and fetal health monitoring	Does not include machine learning-based risk prediction or fall detection.
Ahmed and Kashem, 2020 [5]	Machine Learning, IoT	Maternal health data, IoT communication	Maternal health risk prediction using an IoT-based approach	Does not integrate fall detection, GPS tracking, or remote doctor interaction.
Priyanka et al., 2021 [13]	IoT monitoring	Blood pressure, heart rate, temperature	Remote pregnancy vital-sign monitoring and alert generation	Does not combine ML-based prediction with fall detection.
Liu et al., 2024 [8]	Wearable sensing and AI	Wearable sensors, physiological signals	Review of wearable sensors, data processing, and AI for pregnancy monitoring	Focuses mainly on monitoring technologies rather than an integrated emergency-response platform.
Simegn and Degu, 2025 [20]	Machine Learning	Clinical and maternal health data	Classification of pregnancy-related health risks	Does not provide real-time IoT monitoring or fall detection.
Assaduzzaman et al., 2023 [26]	RF, DT, XGBoost, GB	Maternal Health Risk dataset	Evaluation of multiple ML algorithms for maternal health risk classification	Focuses on risk prediction without integrated real-time IoT and emergency detection.
Harrou et al., 2017 [6]	Computer Vision	Camera-based visual data	Automatic human fall detection using vision-based analysis	Designed for general fall detection and does not address maternal healthcare monitoring.
Seflek and Barstugan, 2026 [38]	YOLO-based detection	Radar / Micro-Doppler data	Investigation of YOLO architectures for human fall detection	Focuses on fall detection and does not integrate maternal health monitoring.
Hu et al., 2026 [4]	YOLO, Edge AI	60-GHz mmWave radar	Real-time vital monitoring and fall detection using edge AI	Radar-based approach without a complete maternal healthcare platform.
Gomes et al., 2026 [7]	Deep Learning, Computer Vision	Vision-based fall detection studies	Systematic mapping of computer vision-based fall detection methods	Provides a review of fall detection approaches rather than an integrated maternal monitoring system.
Proposed Materna AI	Random Forest, YOLOv8n, IoT	ESP32, physiological sensors, camera, GPS, cloud	Integrated maternal monitoring, risk prediction, fall detection, GPS tracking, emergency alerts, and remote healthcare access	Needs clinical testing, calibration, and real-world validation

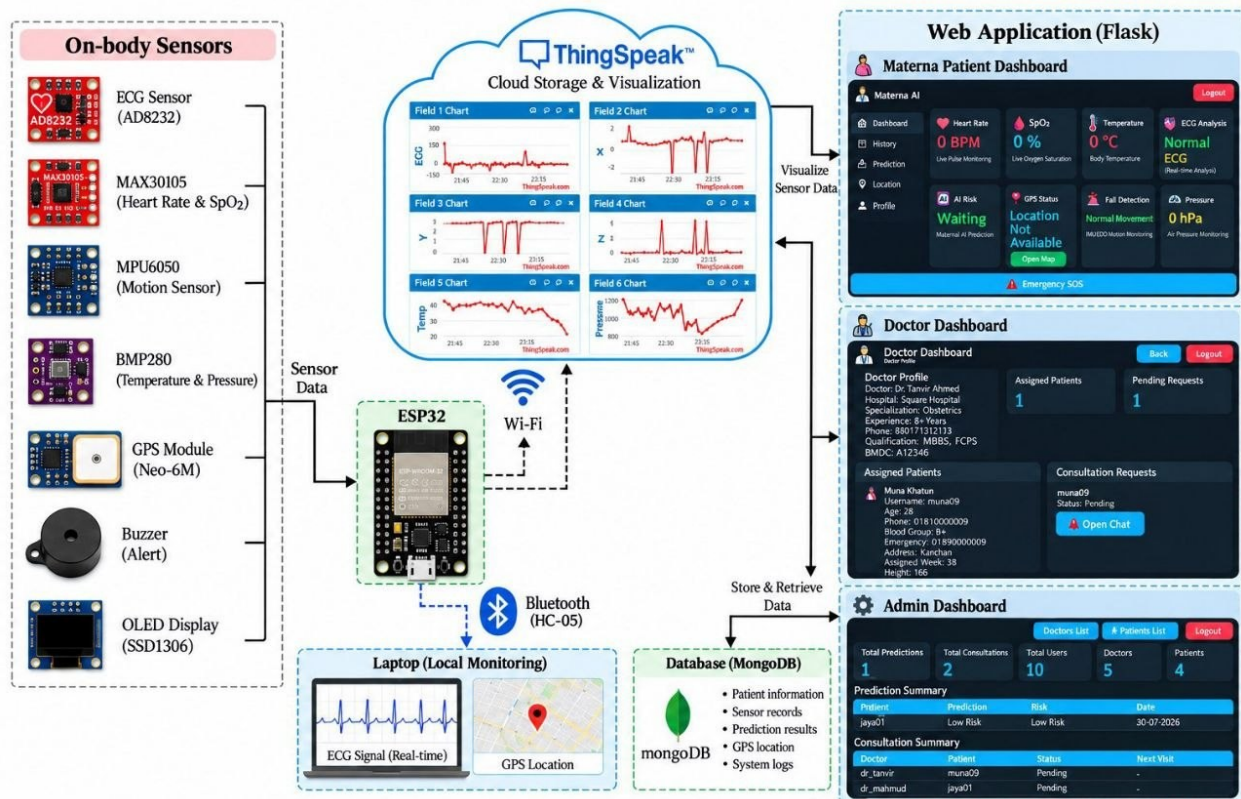
Objectives:

- ✓ To develop an IoT-based system for continuous real-time maternal health monitoring.
- ✓ To predict maternal health risk using machine learning.
- ✓ To detect fall events using YOLO-based computer vision.
- ✓ To provide GPS-based emergency location tracking and alerts.
- ✓ To enable remote monitoring through a cloud and web-based platform.

Circuit diagram of the proposed Materna AI hardware system:



System architecture of the proposed Materna AI system:

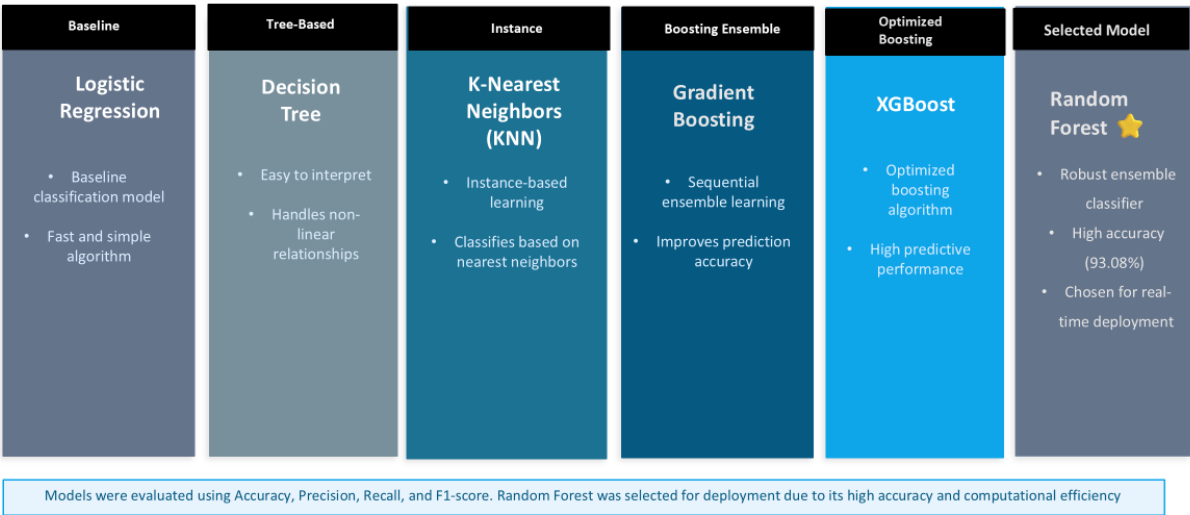


Estimated implementation cost of the proposed Materna AI system:

Hardware Component	Estimated Cost (USD)
ESP32 Development Board	4.27
AD8232 ECG Sensor	9.15
MAX30105 Sensor	7.52
BMP280 Sensor	1.83
MPU6050 Sensor	1.87
GPS Module	4.07
SSD1306 OLED Display	2.44
HC-05 Bluetooth Module	2.68
Buzzer	0.28
Breadboard and Jumper Wires	2.44
18650 Li-ion Battery	2.60
Estimated Total	\$39.15

AI/ML Method:

Machine Learning Models Used



Random Forest achieved an accuracy of 93.08% and was chosen as the final model used for predicting maternal health risk.

Model Performance Comparison

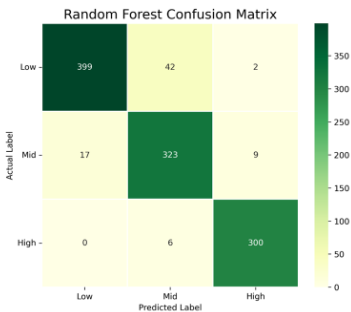
Model	Accuracy	Precision	Recall	F1 Score
Random Forest★	93.08	0.93	0.94	0.93
Logistic Regression	63.84	0.62	0.63	0.61
Decision Tree	92.99	0.93	0.94	0.93
K-Nearest Neighbors (KNN)	90.89	0.91	0.91	0.91
Gradient Boosting	87.70	0.88	0.88	0.88
XGBoost	91.62	0.92	0.92	0.92

Confusion matrix of the Random Forest model for maternal health risk classification:

The confusion matrix was used to check how well the Random Forest model did when it came to classifying the three types of health risks. These types are called Low Risk, Mid Risk and High Risk. The confusion matrix shows how the models predictions compared to the results for each category. It also shows where the model made mistakes.

Performance Evaluation & Confusion Matrix

Class	Precision	Recall	F1-Score	Support
Low Risk	0.96	0.90	0.93	443
Mid Risk	0.87	0.93	0.90	349
High Risk	0.96	0.98	0.97	306
Macro Average	0.93	0.94	0.93	1098
Weighted Average	0.93	0.93	0.93	1098

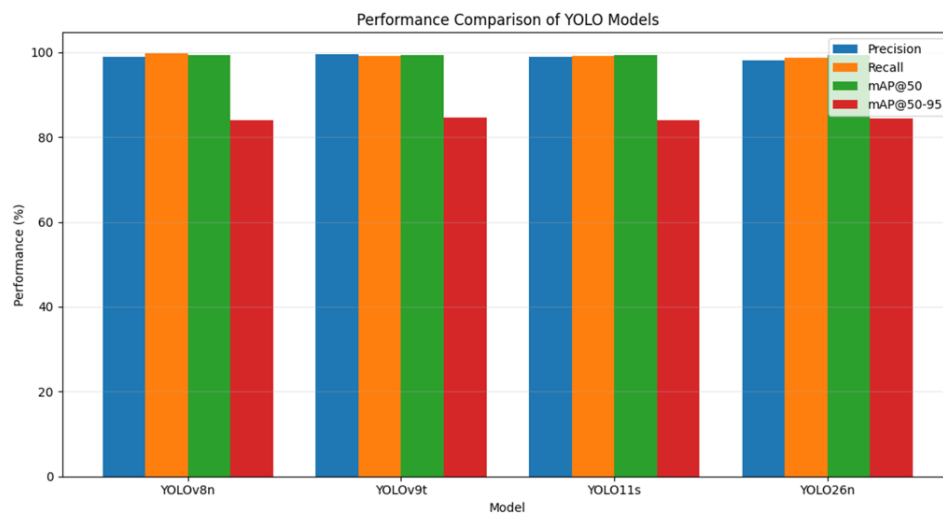


Fall Detection Dataset & Preprocessing

6,446 Total Images	2 Classes	53:13:33 Train:Validation:Test Split
Dataset Overview Source: Roboflow – Human Fall Detection (relabel), Version 1 Classes: Fall, Non-Fall Image Resolution: 640 × 640 Training Images: 3,435 (53.29%) Validation Images: 859 (13.33%) Testing Images: 2,152 (33.39%)		Dataset & Preprocessing ➤ YOLO-format bounding-box annotations ➤ Class 0: Fall, Class 1: Non-Fall ➤ Train/validation/test split ➤ Image resizing to 640 × 640 ➤ Data preparation for YOLO training

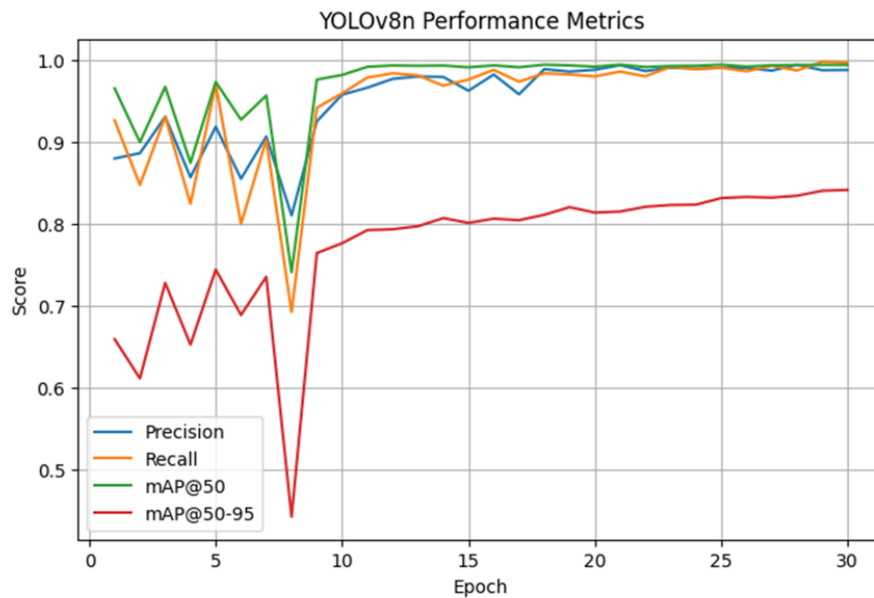
Four YOLO models, YOLOv8n, YOLOv9t, YOLO11s and YOLO26n were evaluated for fall detection using the dataset. Their precision, recall, mAP@50 and mAP@50-95 values were compared

YOLO Model Comparison



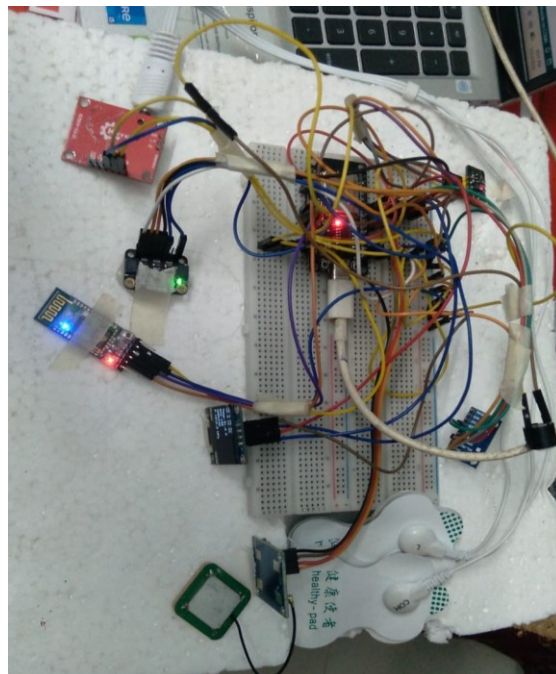
The training performance of the selected YOLOv8n model was checked over 30 epochs. The evaluation used precision, recall, mAP@50 and mAP@50-95.

YOLOv8n Training Performance



Results and Performance:

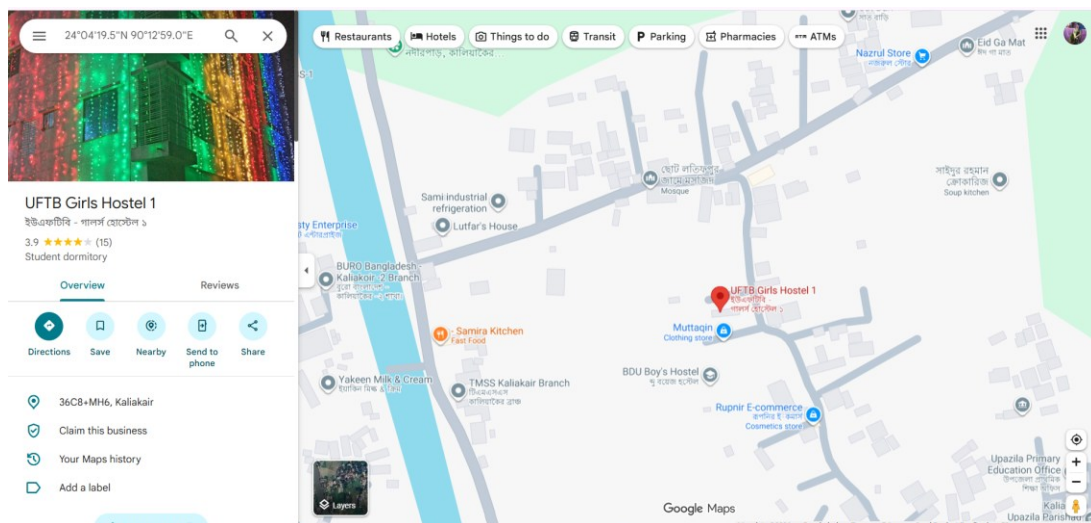
1. Successfully monitored **Heart Rate, SpO₂, ECG, Temperature, Pressure, and Body Movement** in real time.



2. Sensor data were transmitted to the **ThingSpeak cloud platform** for continuous remote monitoring.



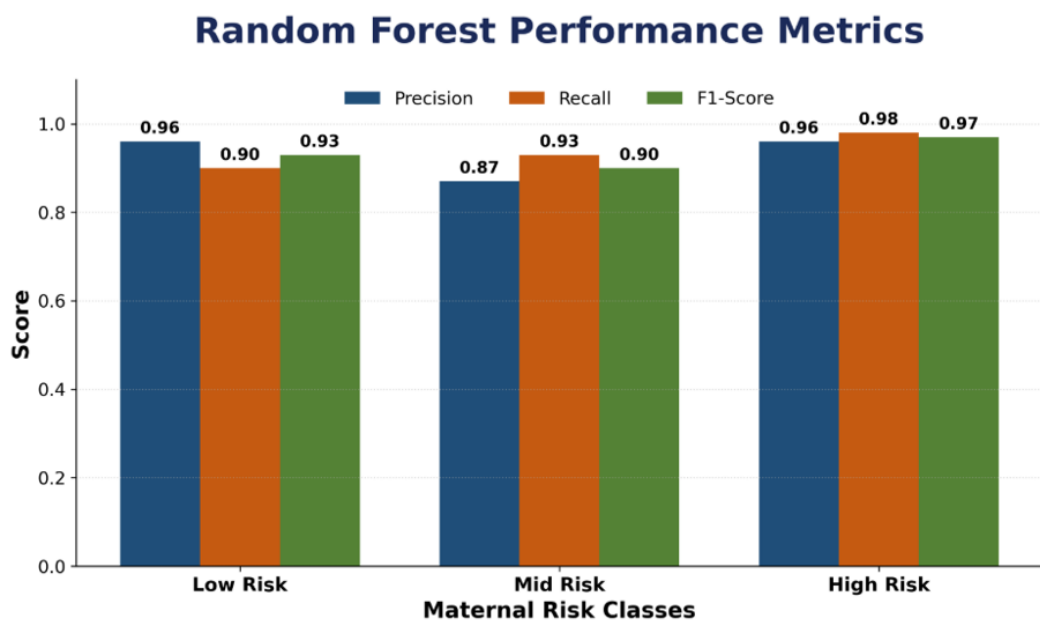
3. The **GPS module** accurately tracked the mother's live location and displayed it on **Google Maps**.



4. The **bluetooth communication** using an HC-05 Bluetooth module and a terminal application, on a smartphone. The ESP32 successfully connected to the HC-05 module. Transmitted ECG data wirelessly. I watched the ECG data stream come through the terminal without any delay. The received data included the ECG value and a Google Maps link generated from the GPS coordinates.



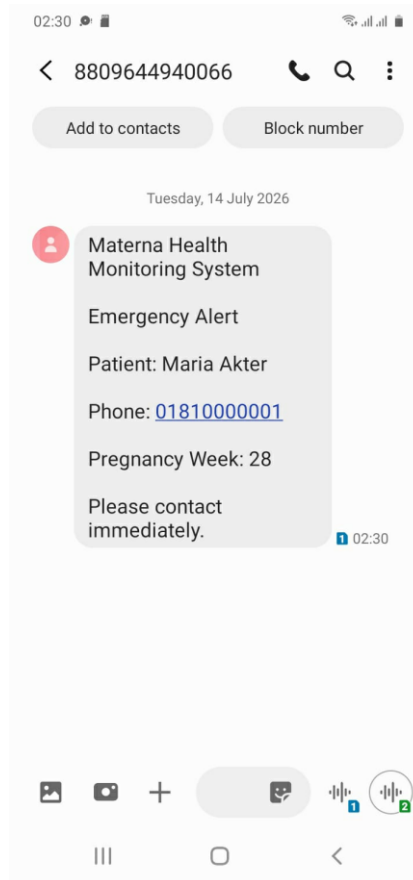
5. Achieved **93.08%** prediction accuracy using the Random Forest model.



6. The **fall detection system** uses a webcam to watch the patient's movements. The user can stop the fall detection system using the provided buttons. The fall detection system shows the current movement status, such, as normal movement. Helps identify possible fall incidents.



7. The Materna AI system offers a way to send alerts when there is a serious problem with a mothers health. If a dangerous situation or a fall happens the system creates a message with the patients details and information, about the pregnancy. This message is sent to the doctors phone number so the doctor can see it is an emergency and do what needs to be done.



Conclusion:

The proposed Materna AI system offers a combined IoT and AI solution for keeping track of health predicting risks detecting falls and offering help in emergencies. The system uses ESP32 along with sensors to gather key data about the body, surroundings, movement and where someone is at any time for ongoing monitoring. The Random Forest model reached an accuracy of 93.08 percent when classifying health risks into Low, Medium and High levels. Also YOLOv8n worked well for detecting falls in time taking 2.3 milliseconds per image. The system also includes warnings based on ECG through a buzzer, tracking location using GPS, wireless communication and a web-based dashboard for watching over someone from a distance. In total bringing together sensors machine learning, fall detection and emergency alerts creates a single platform to help with continuous care, for mothers. The created prototype shows how these technologies can be used together to spot health issues quickly and offer support from away.